# Unit 1　Power Transformer

## Text A　Construction and Principles of Power Transformer

Transformer is an indispensable component in electric energy conversion systems.It makes possible electric generation at the most economical generator voltage，power transfer at the most economical transmission voltage，and power utilization at the most suitable voltage for the particular utilization device.The transformer is also widely used in low-power， lowcurrent electronic and control circuits for performing such functions as matching the impedances of a source and its load for maximum power transfer，isolating one circuit from another，or isolating direct current（DC）while maintaining alternating current（AC）continuity between two circuits.

Essentially，a transformer consists of two or more windings coupled by mutual magnetic flux.If one of these windings，the primary，is connected to an alternating voltage source，an alternating flux will be produced whose amplitude will depend on the primary voltage，the frequency of the applied voltage，and the number of turns.The mutual flux will link the other winding，the secondary，and will induce a voltage in it whose value is depended on the number of secondary turns as well as the magnitude of the mutual flux and the frequency.By properly proportioning the number of primary and secondary turns， almost any desired voltage ratio， or ratio of transformation， can be obtained.

The essence of transformer action requires only the existence of time-varying mutual flux linking two windings.Such action can occur for two windings coupled through air，but coupling between the windings can be made much more effectively using a core of iron or other ferromagnetic material，because most of the flux is then confined to a definite，highpermeability path linking the windings.Such a transformer is commonly called an iron-core transformer.Most transformers are of this type.The following discussion is concerned almost wholly with iron-core transformers.

In order to reduce the losses caused by eddy currents in the core，the magnetic circuit usually consists of a stack of thin laminations.Two common types of construction are shown schematically in Fig.1.1.In the coretype［Fig.1.1（a）］construction，the windings are wound around two legs of a rectangular magnetic core；in the shell-type［Fig.1.1（b）］construction，the windings are wound around the center leg of a three legged core.Siliconsteel laminations of 0.35 mm thickness are generally used for transformers operating at frequencies below a few hundred Hz.Silicon steel has the desirable properties of low cost，low core loss，and high permeability at high flux densities（1.0 to 1.5T）.The cores of small transformers used in communication circuits at higher frequencies and lower energy levels are sometimes made of compressed powdered ferromagnetic alloys known as ferrites.

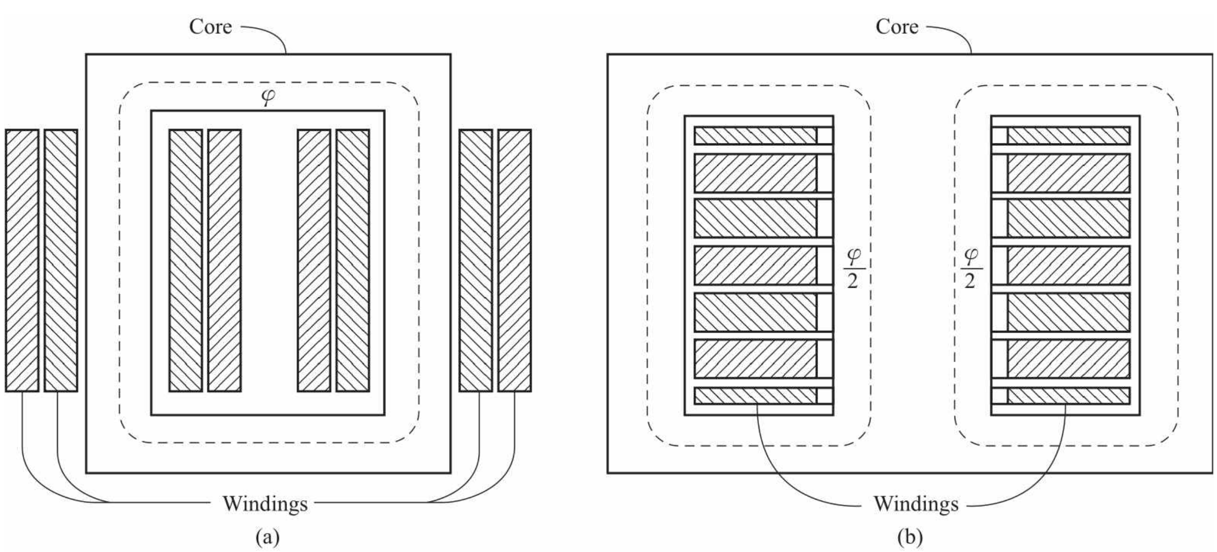



Fig.1.1　Schematic views of transformers

（a）core-type construction；（b）shell-type construction

In both configurations， most of the flux is confined to the core and linking both windings.The windings also produce an additional flux， known as leakage flux， which links one winding without linking the other.Although leakage flux is small fraction of the total flux， it plays an important role in determining the behavior of the transformer.In practical transformers， leakage is reduced by subdividing the windings into sections placed as close together as possible.In the core-type construction，each winding consists of two sections，one section on each of the two legs of the core， the primary and secondary windings being concentric coils.In the shell-type construction， variations of the concentric-winding arrangement may be used， or the windings may consist of a number of thin pancake coils assembled in a stack with primary and secondary coils interleaved.

## Words and Phrases

power transformer　电力变压器

indispensable　*adj*.不可缺少的

power transfer　电能传输，功率转换

isolate　*v*.隔离，孤立，绝缘

matching the impedance　阻抗匹配

winding　*n*.绕组

construction　*n*.结构

primary　*adj*.一次测的

secondary　*adj*.二次测的

turn　*n*.匝数

time-varying　*adj*.时变的

stack　*n*.堆，叠

eddy current　涡流

lamination　*n*.薄层，叠片

silicon-steel laminations　硅钢片

couple　*v*.耦合

loss　*n*.损耗

leakage flux　漏磁通

core type　芯式

shell type　壳式

core　*n*.铁心

wound　*v*.（wind的过去分词）缠，绕

powdered　*adj*.粉末状的，弄成粉的

ferromagnetic　*adj*.铁磁的

alloy　*n*.合金

permeability　*n*.磁导率

interleaved　*adj*.交叉放置的

concentric coils　同心式线圈

pancake coil　扁平线圈

legs of the core　铁心柱

## Notes

1.Such action can occur for two windings coupled through air，but coupling between the windings can be made much more effectively using a core of iron or other ferromagnetic material，because most of the flux is then confined to a definite，high-permeability path linking the windings.

这样的作用也可以发生在通过空气耦合的两个绕组中，但用铁心或其他铁磁材料可以使绕组间的耦合作用更强，因为大部分磁通被限制在与两个绕组交链的高磁导率的路径中。

2.In the core type the windings are wound around two legs of a rectangular magnetic core［Fig.1.1（a）］；in the shell type the windings are wound around the center leg of a three-legged core［Fig.1.1（b）］.

芯式变压器的绕组绕在两个矩形铁心柱上［图1.1（a）］。壳式变压器的绕组绕在铁心柱的中间心柱上［图1.1（b）］。

## Exercises

### 1.Answer the following questions according to the text

（1）How many types of core construction do power transformers have？

（2）Why does power transformer consist of a stack of thin laminations？

（3）What does the alternating flux amplitude depend on？

（4）What is leakage flux？

（5）In order to reduce the losses caused by eddy currents in the core，Silicon-steel laminations are generally used for transformers operating at frequencies below a few hundred Hz.Right or wrong？

### 2.Translate the following sentences into Chinese according to the text

（1）The transformer is also widely used in low-power，low-current electronic and control circuits for performing such functions as matching the impedances of a source and its load for maximum power transfer.

（2）By properly proportioning the number of primary and secondary turns，almost any desired voltage ratio，or ratio of transformation，can be obtained.

（3）The windings also produce additional flux，known as leakage flux，which links one winding without linking the other.

（4）In the core-type construction，each winding consists of two sections，one section on each of the two legs of the core，the primary and secondary windings being concentric coils.

### 3.Translate the following paragraph into Chinese

In addition to the various power transformers，two special-purpose transformers are used with electric machinery and power systems.The first of these special transformers is a device specially designed to sample a high voltage and produce a low secondary voltage directly proportional to it.Such a transformer is a potential transformer（电压互感器）.A power transformer also produces a secondary voltage directly proportional to its primary voltage；the difference between a potential transformer and a power transformer is that the potential transformer is designed to handle only a very small current.The second type of special transformer is a device designed to provide a secondary current much smaller than but directly proportional to its primary current.This device is called a current transformer（电流互感器）.

# Unit 4　Permanent Magnet（PM）Machines

## Text A　Introduction to Permanent Magnet（PM）Machines

### 1.Introduction

With the emergence of modern ferrite and rare earth magnets，permanent magnet（PM）machines are being increasingly used in applications where efficiency，or more importantly，losses are of concern.One such application is in servo drives in which the motor is commanded to operate for a long period at，or very near，zero speed.In such applications，the heat rejection of a conventional squirrel cage induction motor drive could be insufficient to keep the machine（usually the rotor）at tolerable levels.Since PM machines use magnets to supply the excitation current（current required to magnetize the iron parts），a substantial reduction of the stator current is possible，thereby greatly reducing the losses.Increased use of such machines will inevitably also become important in conventional operation from the utility supply when the cost of energy increases to a sufficiently high value.

The use of permanent magnets（PMs）in construction of electrical machines brings the following benefits:

➢No electrical energy is absorbed by the field excitation system and thus there are no excitation losses which means substantial increase in efficiency.

➢Higher power density and/or torque density than when using electromagnetic excitation.

➢Better dynamic performance than motors with electromagnetic excitation（higher magnetic flux density in the air gap）.

➢Simplification of construction and maintenance.

➢Reduction of prices for some types of machines.

### 2.Classification of permanent magnet electric motors

In general，rotary PM motors for continuous operation are classified into DC brush commutator motors，DC brushless motors and AC synchronous motors.The construction of a PM DC commutator motor is similar to a DC motor with the electromagnetic excitation system replaced by PMs.PM DC brushless and AC synchronous motor designs are practically the same:with a polyphaser stator and PMs located on the rotor.The only difference is in the control and shape of the excitation voltage:an AC synchronous motor is fed with more or less sinusoidal waveforms which in turn produce a rotating magnetic field.In PM DC brushless motors the armature current has a shape of a square（trapezoidal）waveform，only two phase windings（for Y connection）conduct the current at the same time，and the switching pattern is synchronized with the rotor angular position（electronic commutation）.

The armature current of synchronous and DC brushless motors is not transmitted through brushes，which are subject to wear and require maintenance.Another advantage of the brushless motor is the fact that the power losses occur in the stator，where heat transfer conditions are good.Consequently the power density can be increased as compared with a DC commutator motor.In addition，considerable improvements in dynamics can be achieved because the air gap magnetic flux density is high，the rotor has a lower inertia and there are no speed-dependent current limitations.Thus， the volume of a brushless PM motor can be reduced by 40%to 50%while still keeping the same rating as that of a PM commutator motor（Fig.1.8）.

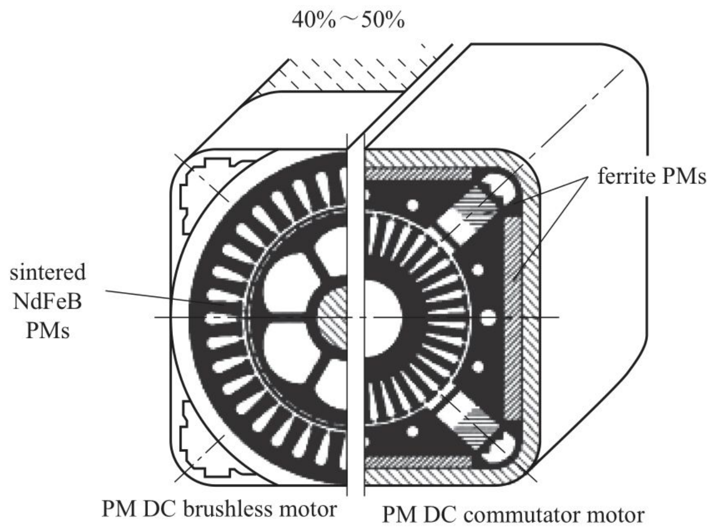


Fig.1.8　Comparison of PM brushless and PM DC commutator motors.

## 3.Applications of permanent magnet motors

PM motors are used in a broad power range from m W to hundreds k W.There are also attempts to apply PMs to large motors rated at minimum 1 MW.Thus，PM motors cover a wide variety of application fields，from stepping motors for wrist watches， through industrial drives for machine tools to large PM synchronous motors for ship propulsion（navy frigates，cruise ships，medium size cargo vessels and ice breakers）.The automotive industry is the biggest user of PM DC commutator motors.The number of auxiliary DC PM commutator motors can vary from a few in an inexpensive car to about one hundred in a luxury car.PM brushless motors rated from 50 to 100 k W seem to be the best propulsion motors for electric and hybrid road vehicles.

## Words and Phrases

ferrite　*n*.铁氧体

rare earth magnet　稀土磁体

servo drive　伺服驱动系统

squirrel cage induction motor drive　笼型感应电机驱动系统

tolerable　*adj*.可忍受的；过得去的

magnetize　*adj*.使……磁化；使……有磁性

substantial　*adj*.大量的；结实的，牢固的；重大的

inevitably　*adv*.不可避免地，自然而然地；必然地，无疑的

conventional　*adj*.传统的；习用的，平常的；依照惯例的；约定的

utility　*n*.功用，效用；有用的物体或器械；

*adj*.有多种用途的；各种工作都会做的

power/torque density　功率、转矩密度

electromagnetic　*adj*.电磁的

dynamic performance　动态性能

flux　*n*.（磁）通量

rotary　*adj*.旋转的

DC brush commutator motors　直流有刷换向电动机

DC brushless motors　直流无刷电动机

AC synchronous motors　交流同步电动机

sinusoidal　*adj*.正弦的

trapezoidal　*adj*.梯形的

be subject to　受……支配；从属于……

considerable　*adj*.相当大（或多）的

inertia　*n*.惯性；惰性

auxiliary　*adj*.辅助的；备用的，补充的；附加的；副的

hybrid　*adj*.混合的

## Notes

1.With the emergence of modern ferrite and rare earth magnets，permanent magnet（PM）machines are being increasingly used in applications where efficiency，or more importantly，losses are of concern.

随着现代铁氧体和稀土磁体的出现，永磁电机正越来越多地应用于更关注效率或损耗的场合中。

2.Increased use of such machines will inevitably also become important in conventional operation from the utility supply when the cost of energy increases to a sufficiently high value.

当能源成本很高时，在传统操作中更多地使用这类电机必然变得很重要。

3.The only difference is in the control and shape of the excitation voltage:an AC synchronous motor is fed with more or less sinusoidal waveforms which in turn produce a rotating magnetic field.

唯一的区别是励磁电压的控制和形状:交流同步电动机采用正弦波或近似正弦波供电，这些波形进而产生旋转磁场。

## Exercises

### 1.Answer the following questions according to the text

（1）The excitation end does not consume electric energy for permanent magnet motors.True or false？

（2）What is in common about the structure design of PM DC brushless and AC synchronous motor？

（3）What is difference between PMDC brushless and AC synchronous motor？

（4）Why is the power density of the brushless motor higher than that of DC commutator motor？

（5）What kind of motoris considered as the best driving motor for electric and hybrid vehicles？

## 2.Translate the following sentences into Chinese according to the text

（1）One such application is in servo drives in which the motor is commanded to operate for a long period at，or very near，zero speed.

（2）The construction of a PM DC commutator motor is similar to a DC motor with the electromagnetic excitation system replaced by PMs.

（3）The armature current of synchronous and DC brushless motors is not transmitted through brushes，which are subject to wear and require maintenance.

（4）In addition， considerable improvements in dynamics can be achieved because the air gap magnetic flux density is high， the rotor has a lower inertia and there are no speeddependent current limitations.

（5）The number of auxiliary DC PM commutator motors can vary from a few in an inexpensive car to about one hundred in a luxury car.

## 3.Translate the following paragraph into Chinese

Rare earth PMs can not only improve the motor's steady state performance but also the power density（output power-to-mass ratio），dynamic performance， and quality.The prices of rare earth magnets are also dropping，which is making these motors more popular.The improvements made in the field of semiconductor drives have meant that the control of brushless motors has become easier and cost effective， with the possibility of operating the motor over a large range of speeds while still maintaining good efficiency.

# Unit 6　Electrical Apparatus Reliability

## Test A　Electrical Apparatus Reliability Tests

Electrical apparatus reliability is referred to the apparatus'ability to achieve the prescribed functions within prescribed time under prescribed conditions.It reflects an apparatus'good performances in failure-free，maintainability，durability，validity，economy-utility and longevity.Thus，reliability is an important quality index.

Apparatus reliability data can be achieved through tests in lab or field survey.In recent 20 years，such reliability test researches in China are almost confined in labs.The products that have been carried out such tests involve electromagnetism auxiliary relay，small load AC contactor，fuse，circuit-breaker，electric leakage switch，and so on.Such researches are effective to improve low voltage apparatus quality.Carrying such reliability tests in lab has some advantages:such as the same test conditions for test product，the same failure judgment standards，which makes it easier to compare the test results，and determine the proper reliability index and judging method.But the testing conditions are not the same with product using conditions.The reliability tests of the main technique performances for some products cost too much.And for the limitations of expenses and the number of some valuable products，it is almost impossible to do complete reliability tests.For example，the reliability tests for contactor only test the main circuits and non-load operating performance of some small load AC contactor.And the tests of longevity and make-break capability can not be carried out due to the high expenses.The reliability tests of fuse only can test the currenttime influences on aging of fuse caused by the starting current of electromotor.And the tests on its breaking capability can not be carried out.The reliability tests of breakers only can test the operations of non-load release.And as for electricity protective characteristics，longevity and utmost breaking capability are not involved.

Electrical apparatus operate under different operating environments.The reliability test in lab can not consider all possible conditions including temperature，humidity，impact，vibration，harmful media etc.Whereas for field survey，through choosing the location of field survey，the reliability under all kinds of practical conditions can be reflected clearly.The reliability design of a set of low-voltage electrical apparatus needs complete reliable data of all components，which would be limited to achieve in lab.But through field survey，the tests can reflect the practical operating circumstances and provide reliable data for design.

The traditional apparatus reliability tests carried out in lab have some limitations，such as too strict test conditions or so less outputs.For the valuable products，to carry reliability tests at the operating location is much better than in the lab.To ensure the high reliability during the use of such apparatus and to avoid losses taken by maintaining or changing parts after failure appears，collecting and analyzing the data of the apparatus reliability indexes during its normal use become very important.Mastering the invalidate data and then giving precaution before apparatus fails are necessary.

Biology Immune System is a complex system composed by organs，cells，and molecules.Except for nerve system it's another one that the organism can recognize the stimulation from"itself"and"non-itself"，response precisely and reserve the memory.It has the characteristics such as pattern recognition，self-learning，memory achieving，distributed detecting and so on.Artificial immune system，which is developed from biology immune system，can recognize and process the input information，learn the right input，produce immunity and respond to the abnormal input.Taking the reliability data collected from the spot as the input for artificial immune system，the system can judge the abnormal operating state of apparatus.These make it possible to maintain or change the parts of electrical apparatus in time and avoid losses.So，artificial immune system is a novel method of electrical apparatus reliability test.

## Words and Phrases

electrical apparatus　电器

reliability　*n.*可靠性

failure-free　*n.*无故障性

maintainability　*n.*维修性

durability　*n.*耐久性

validity　*n.*有效性

economy-utility　*n.*使用经济性

index　*n.*指标

fuse　*n.*熔断器

field survey　现场调查

electromagnetism auxiliary relay　电磁式中间继电器

small load AC contactor　小容量交流接触器

electric leakage switch　漏电开关

circuit-breaker　*n.*断路器

electromotor　*n.*电动机

longevity　*n.*寿命

make-break capability　分断能力

humidity　*n.*湿度

impact　*n.*冲击，撞击

vibration　*n.*振动

harmful media　有害介质

aging　*n.*老化

precaution　*n.*预警，防范

spot　*n.*现场

biology immune　生物免疫

nerve system　神经系统

characteristic　*n.*特性，特征

pattern recognition　模式识别

self-learning　*n.*自学习

distributed detecting　分布式检测

abnormal　*adj.*反常的

novel　*adj.*新颖的

## Notes

1.Electrical apparatus reliability is referred to the apparatus ability to achieve the prescribed functions within prescribed time under prescribed conditions.

电器的可靠性是指电器产品在规定的条件下和规定的时间内完成规定功能的能力。

2.Carrying such reliability tests in lab has some advantages:such as the same test conditions for test product，the same failure judgment standards，which make it easier to compare the test results，determine the proper reliability index and judge method.

在实验室内进行可靠性试验研究，具有试品的试验条件和故障评判标准统一、可比性强以及便于确定可靠性考核指标和评定方法的优点。

3.Except for nerve system it's another one that the organism can recognize the stimulation from"itself"and"non-itself"，response precisely and reserve the memory.

它是除神经系统外，机体能特异地识别“自己”和“非己”刺激，对之做出精确应答并保留记忆的功能系统。

## Exercises

### 1.Answer the following questions according to the text

（1）What is electrical apparatus reliability？

（2）Apparatus reliability data can only be achieved through tests in lab.Right or wrong？

（3）Where the traditional apparatus reliability tests carried out in？

（4）Biology Immune System has the characteristics such as pattern recognition，selflearning，memory achieving，distributed detecting.Right or wrong？

（5）Artificial immune system is one of the methods of electrical apparatus reliability test.Right or wrong？

### 2.Translate the following sentences into Chinese according to the text

（1）The products that have been carried out such tests involve electromagnetism auxiliary relay，small load AC contactor，fuse，circuit-breaker，electric leakage switch，and so on.

（2）The reliability tests of the main technique performances for some products costs too much.

（3）And for the limitations of expenses and the number of some valuable products，it is almost impossible to do complete reliability tests.

（4）The reliability test in lab can not consider all possible conditions including temperature，humidity，impact，vibration，harmful media etc.

（5）Mastering the invalidate data and then giving precaution before apparatus fails are necessary.

### 3.Translate the following paragraph into Chinese

In order to confirm and guarantee the reliability of electrical contacts，a lot of experience is required，especially on how to develop additional testing methods on top of the requirements given though national and international standards.It is very important that the manufacturers of devices with electrical contacts are developing such methods（temperature rise tests， short circuit tests， high voltage tests， etc.）and continuously are using them as routine tests in order to confirm the reliability of electrical contacts.

# Unit 1　Magnetism and Electromagnetism

## Text A　Electromagnetism

The discovery of the relationship between magnetism and electricity， like many other scientific discoveries， was stumbled upon almost by accident.The Danish physicist Hans Christian Oersted was lecturing one day in 1820 on the possibility of electricity and magnetism being related to one another，and in the process demonstrated it conclusively by experiment in front of his whole class！ By passing an electric current through a metal wire suspended above a magnetic compass，Oersted was able to produce a definite motion of the compass needle in response to the current.What began as conjecture at the start of the class session was confirmed as a fact at the end.His serendipitous discovery paved the way for a whole new branch of science:electromagnetics.

Detailed experiments showed that the magnetic field produced by an electric current is always oriented perpendicular to the direction of flow.A simple method of showing this relationship is called the right-hand rule.Simply stated，the right-hand rule says that the magnetic flux lines produced by a current-carrying wire will be oriented the same direction as the curled fingers of a person's right hand，with the thumb pointing in the direction of electron flow，which is shown in Fig.2.1.

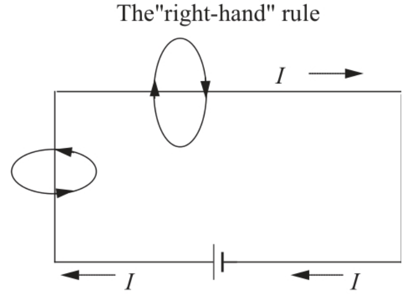



Fig.2.1　Magnetic field produced by an electric current

The magnetic field encircles this straight piece of current-carrying wire， the magnetic flux lines having no definite"north"or"south"poles.

While the magnetic field surrounding a current-carrying wire is indeed interesting，it is quite weak for common amounts of current，able to deflect a compass needle and not much more.To create a stronger magnetic field force with the same amount of electric current，we can wrap the wire into a coil shape，where the circling magnetic fields around the wire will join to create a larger field with a definite magnetic（north and south）polarity.See Fig.2.2.

The amount of magnetic field force generated by a coiled wire is proportional to the current through the wire multiplied by the number of"turns"or"wraps"of wire in the coil.This field force is called magnetomotive force（MMF），and is very much analogous to electromotive force（EMF）in an electric circuit.

An electromagnet is a piece of wire intended to generate a magnetic field with the passage of electric current through it.Though all current-carrying conductors produce magnetic fields， an electromagnet is usually constructed in such a way as to maximize the strength of the magnetic field it produces for a special purpose.Electromagnets find frequent application in research，industry，medical，and consumer products.

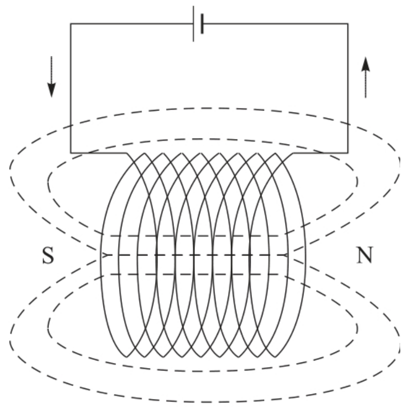


Fig.2.2　Circling magnetic fields around the wire

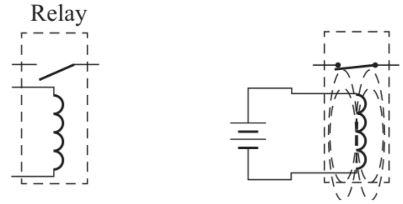


Fig.2.3　Applying current through the coil causes the switch to close

As an electrically-controllable magnet， electromagnets find application in a wide variety of"electromechanical"devices:machines that effect mechanical force or motion through electrical power.Perhaps the most obvious example of such a machine is the electric motor.Another example is the relay， an electrically-controlled switch.If a switch contact mechanism is built so that it can be actuated（opened and closed）by the application of a magnetic field，and an electromagnet coil is placed in the near vicinity to produce that requisite field，it will be possible to open and close the switch by the application of a current through the coil.In effect，this gives us a device that enables electricity to control electricity.See Fig.2.3.

## Words and Phrases

Magnetism　*n.*磁学

electricity　*n.*电学

stumble upon　偶然发现

Danish　*adj.*丹麦的

Oersted　*n.*奥斯特（人名）

one another　彼此，互相

compass needle　罗盘针

suspend　*v.*吊，悬挂，悬浮

compass　*n.*罗盘

electric current　电流

conjecture　*n.*推测，猜想

serendipitous　*adj.*偶然发现的

electromagnetics　*n*.电磁学

detailed experiment　详细的试验

magnetic field　磁场

perpendicular　*adj*.垂直的，正交的

right-hand rule　右手定则

encircle　*v*.环绕，围绕

magnetic flux line　磁力线

current-carrying wire　载流导线

magnetomotive force　磁动势

electromotive force　电动势

electric motor　电动机

electrically-controlled switch　电控开关

## Notes

1.By passing an electric current through a metal wire suspended above a magnetic compass，Oersted was able to produce a definite motion of the compass needle in response to the current.

奥斯特在悬放于磁罗盘上方的金属线中通入电流，发现罗盘针有明确的运动。

2.The right-hand rule says that the magnetic flux lines produced by a current-carrying wire will be oriented the same direction as the curled fingers of a persons' right hand，with the thumb pointing in the direction of electron flow.

右手定则:当大拇指指向电流的方向时，载流导线所产生的磁力线的方向与人的右手四指的弯曲方向一致。

3.The amount of magnetic field force generated by a coiled wire is proportional to the current through the wire multiplied by the number of"turns"or"wraps"of wire in the coil.

导线线圈产生的磁动势的大小与导线中流过的电流和线圈匝数的乘积呈正比。

## Exercises

### 1.Answer the following questions according to the text

（1）Who discovered the relationship between magnetism and electricity firstly？

（2）What is the right-hand rule used to explain the relationship between magnetism and electricity？

（3）The magnetic field force produced by a current-carrying wire can be greatly increased by shaping the wire into a coil instead of a straight line.Right or wrong？

（4）What is very much analogous to electromotive force（*E*）in an electric circuit？（5）Where is Electromagnets widely used in？

### 2.Translate the following sentences into Chinese according to the text

（1）His serendipitous discovery paved the way for a whole new branch of science:electromagnetics.

（2）To create a stronger magnetic field force with the same amount of electric current，we can wrap the wire into a coil shape， where the circling magnetic fields around the wire will join to create a larger field with a definite magnetic polarity.

（3）An electromagnet is a piece of wire intended to generate a magnetic field with the passage of electric current through it.

（4）As an electrically-controllable magnet， electromagnets find application in a wide variety of"electromechanical"devices:machines that effect mechanical force or motion through electrical power.

## 3.Translate the following paragraph into Chinese

Electromechanical devices which employ magnetic fields often use ferromagnetic materials for guiding and concentrating these fields.Because the magnetic permeability of ferromagnetic materials can be large（up to tens of thousands times that of the surrounding space），most of the magnetic flux is confined to fairly well-defined paths determined by the geometry of the magnetic material.

# Unit 4　Electromagnetic Interference and Electromagnetic Compatibility

## Text A　Electromagnetic Interference and Electromagnetic Compatibility

Electromagnetic interference（EMI）is a major problem in modern electronic circuits.To overcome the interference，the designer has to either remove the source of the interference，or protect the circuit being affected.The ultimate goal is to have the circuit operating as intended to achieve electromagnetic compatibility（EMC）.

EMI:Electromagnetic emissions from a device or system that interfere with the normal operation of another device or system.

EMC:The ability of a device or system to function without error in its intended electromagnetic environment.

**Examples of EMC problems:**

➢The use of mobile phones on the plane will interfere with the unlimited electrical contact between the plane and the ground control tower.

➢Your car radio buzzes when you drive under a power line.

➢A helicopter goes out of control when it flies too close to a radio tower.

➢Your telephone is damaged by lightning-induced surges on the phone line.

➢Your new memory board is destroyed by an unseen discharge as you install it.

➢Noise of switching power supply.

➢Your laptop computer interferes with your aircrafts' rudder control.

➢The airport radar interferes with your laptop computer display.

There are three essential elements to any EMC problem，which are shown in Fig.2.13.There must be a source of an electromagnetic phenomenon，a receptor（or victim）that cannot function properly due to the electromagnetic phenomenon，and a path between them that allows the source to interfere with the receptor.Each of these three elements must be present although they may not be readily identified in every situation.Electromagnetic compatibility problems are generally solved by identifying at least two of these elements and eliminating（or attenuating）one of them.

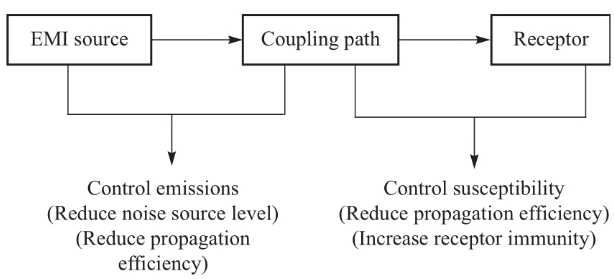



Fig.2.13　EMC elements

For example，in the case of the nuclear power plant，the receptor was readily identified.The turbine control valves were malfunctioning.The source and the coupling path were originally unknown，however an investigation revealed that the walkie talkies used by the plant employees were the source.Although at this point the coupling path was not known，the problem could be solved by eliminating the source（e.g.restricting the use of lowpower radio transmitters in certain areas）.A more thorough and perhaps more secure approach would be to identify the coupling path and take steps to eliminate it.For example，suppose it was determined that radiated emissions from a walkie talkie were inducing currents on a cable that was connected to a printed circuit card that contained a circuit that controlled the turbine valves.If the operation of the circuit was found to be adversely affected by these induced currents，a possible coupling path would be identified.Shielding，filtering，or rerouting the cable，and filtering or redesigning the circuit would then be possible methods of attenuating the coupling path to the point where the problem is non-existent.

The source of the tramway problem was thought to be transients on the tramway's power.The coupling path was presumably through the power supply to the speed control circuit，although investigators were unable to reproduce the failure so the source and coupling path were never identified conclusively.The receptor，on the other hand，was clearly shown to be the speed control circuit and this circuit was modified to keep it from becoming confused by unintentional random inputs.In other words，the solution was to eliminate the receptor by making the speed control circuit immune to the electromagnetic phenomenon produced by the source.

Potential sources of electromagnetic compatibility problems include radio transmitters，power lines，electronic circuits，lightning，lamp dimmers，electric motors，arc welders and just about anything that utilizes or creates electromagnetic energy.Potential receptors include radio receivers，electronic circuits，appliances，people，and just about anything that utilizes or can detect electromagnetic energy.

Methods of coupling electromagnetic energy from a source to a receptor fall into one of four categories.

➢Conducted（electric current）

➢Inductively coupled（magnetic field）

➢Capacitively coupled（electric field）

➢Radiated（electromagnetic field）

Coupling paths often utilize a complex combination of these methods making the path difficult to identify even when the source and receptor are known.There may be multiple coupling paths and steps taken to attenuate one path may enhance another.

## Words and Phrases

electromagnetic interference（EMI） 电磁干扰

electromagnetic compatibility（EMC） 电磁兼容

source of the interference 干扰源

ultimate goal 最终目标

function *v.*运行

interfere with 干扰，干涉

power line 输电线

out of control 失控

emission *n.*发射，辐射

FM *abbr.*调频

CB radio（Citizen's Band radio） 民用波段收音机

stereo *n.*立体声系统

radio tower 无线电接收塔

airport radar 机场雷达

laptop computer 膝上型计算机

helicopter *n.*直升机

VCR *abbr.*录像机

rudder *n.*舵，方向舵

receptor *n.*受体，接收器

coupling path 耦合路径

eliminate *v.*消除

nuclear power plant 核电站

turbine control valve 涡轮控制阀

malfunction *n.*故障

walkie talkies 步话机

shield　*v*.屏蔽

filter　*v*.滤波

reroute　*v*.改变路线

attenuate　*v*.消弱

unintentional　*adj*.无意识的，无心的

radiate　*v*.辐射

take steps　设法，采取措施

inductively coupled　电感性耦合

capacitively coupled　电容性耦合

## Notes

1.Each of these three elements must be present although they may not be readily identified in every situation.Electromagnetic compatibility problems are generally solved by identifying at least two of these elements and eliminating（or attenuating）one of them.

三个要素都一定存在，而在有些情况下它们不容易被识别出来，解决电磁兼容问题一般需要至少识别出两个要素并消除或消弱其中之一。

2.A more thorough and perhaps more secure approach would be to identify the coupling path and take steps to eliminate it.

更彻底而且可能更安全的方法是确定耦合路径并逐步消除之。

3.Potential sources of electromagnetic compatibility problems include radio transmitters，power lines，electronic circuits，lightning，lamp dimmers，electric motors，arc welders and just about anything that utilizes or creates electromagnetic energy.

电磁兼容问题的潜在的干扰源包括无线电发射器、输电线、电子电路、闪电、调光灯、电机、电弧焊接和任何利用或产生电磁能量的东西。

## Exercises

### 1.Answer the following questions according to the text

（1）What is electromagnetic interference（EMI）?

（2）What is electromagnetic compatibility（EMC）?

（3）What are the three essential elements to any EMC problem?

（4）There are four main categories of coupling electromagnetic energy from a source to a receptor.Right or wrong?

（5）Coupling paths often utilize a complex combination of several methods making the path difficult to identify.Right or wrong?

### 2.Translate the following sentences into Chinese according to the text

（1）Electromagnetic interference（EMI）is a major problem in modern electronic circuits.To overcome the interference，the designer has to either remove the source of the interference，or protect the circuit being affected.

（2）There must be a source of an electromagnetic phenomenon，a receptor（or victim）that cannot function properly due to the electromagnetic phenomenon，and a path between them that allows the source to interfere with the receptor.

（3）The turbine control valves were malfunctioning.The source and the coupling path were originally unknown，however an investigation revealed that the walkie talkies used by the plant employees were the source.

（4）Shielding，filtering，or rerouting the cable，and filtering or redesigning the circuit would then be possible methods of attenuating the coupling path to the point where the problem is non-existent.

（5）In other words，the solution was to eliminate the receptor by making the speed control circuit immune to the electromagnetic phenomenon produced by the source.

## 3.Translate the following paragraph into Chinese

Coupling can also occur in circuits that share common impedances.For instance， two circuits that share the conductor carrying the supply voltage and the conductor carrying the return path to ground.If one circuit creates a sudden demand in current， the other circuit's voltage supply will drop due to the common impedance both circuits share between the supply lines and the source impedance.This coupling effect can be reduced by decreasing the common impedance.Coupling also can occur with radiated electric and magnetic fields which are common to all electrical circuits.Whenever current changes， electromagnetic waves are generated.These waves can couple over to nearby conductors and interfere with other signals within the circuit.

# Unit 5 Electrotechnical Materials and the Characteristics

## Text A Introduction of the Electrotechnical Materials

Soft magnetic materials are used in devices that are subjected to alternating magnetic fields and in which energy losses must be low; one familiar example consists of transformer cores.For this reason the relative area within the hysteresis loop must be small; it is characteristically thin and narrow.Consequently, a soft magnetic material must have a high initial permeability and a low coercivity.A material possessing these properties may reach its saturation magnetization with a relatively low applied field and still has low hysteresis energy losses.The saturation field or magnetization is determined only by the composition of the material.For example, in cubic ferrites, substitution of a divalent metal ion such as for in will change the saturation magnetization.However, susceptibility and coercivity which also influence the shape of the hysteresis curve, are sensitive to structural variables rather than to composition.For example, a low value of coercivity corresponds to the easy movement of domain walls as the magnetic field changes magnitude and/or direction.Structural defects such as particles of a nonmagnetic phase or voids in the magnetic material tend to restrict the motion of domain walls, and thus increase the coercivity.Consequently, a soft magnetic material must be free of such structural defects.Another property consideration for soft magnetic materials is electrical resistivity.In addition to the hysteresis energy losses described above, energy losses may result from electrical currents that are induced in a magnetic material by a magnetic field that varies in magnitude and direction with time; these are called eddy currents.It is most desirable to minimize these energy losses in soft magnetic materials by increasing the electrical resistivity.This is accomplished in ferromagnetic materials by forming solid solution alloys; iron-silicon and iron-nickel alloys are examples.The ceramic ferrites are commonly used for applications requiring soft magnetic materials because they are intrinsically electrical insulators.In addition, soft magnetic materials are used in generators, motors, dynamos, and switching circuits.

### 1.Hard Magnetic Materials

Hard magnetic materials are utilized in permanent magnets, which must have a high resistance to demagnetization.In terms of hysteresis behavior, a hard magnetic material has a high remanence, coercivity, and saturation flux density, as well as a low initial permeability, and high hysteresis energy losses.The two most important characteristics relative to applications for these materials are the coercivity and what is termed the"energy product", designated as $BH_{max}$.This $BH_{max}$ corresponds to the area of the largest $B$-$H$ rectangle that can be constructed within the second quadrant of the hysteresis curve.The value of the energy product is representative of the energy required to demagnetize a permanent magnet; that is, the larger $BH_{max}$ the harder is the material in terms of its magnetic characteristics.Again, hysteresis behavior is related to the ease with which the magnetic domain boundaries move; by impeding domain wall motion, the coercivity and susceptibility are enhanced, such that a large external field is required for demagnetization.Furthermore, these characteristics are interrelated to the microstructure of the material.

### 2.Dielectric Materials

A number of ceramics and polymers are utilized as insulators and/or in capacitors.Many of the ceramics, including glass, porcelain, steatite, and mica, have dielectric constants within the range of 6 to 10.These materials also exhibit a high degree of dimensional stability and mechanical strength.Typical applications include power line and electrical insulation， switch bases， and light receptacles.The titania（$TiO_2$）and titanate ceramics， such as barium titanate（$BaTiO_3$）， can be made to have extremely high dielectric constants， which render them especially useful for some capacitor applications.The magnitude of the dielectric constant for most polymers is less than for ceramics，since the latter may exhibit greater dipole moments:values for polymers generally lie between 2 and 5.These materials are commonly utilized for insulation of wires， cables， motors， generators， and so on， and， in addition，for some capacitors.

## 3.Conductive Materials

Most metals are extremely good conductors of electricity because of the large numbers of free electrons that have been excited into empty states above the Fermi energy.At this point it is convenient to discuss conduction in metals in terms of the resistivity，the reciprocal of conductivity；the reason for this switch in topic should become apparent in the ensuing discussion.Since crystalline defects serve as scattering centers for conduction electrons in metals， increasing their number raises the resistivity（or lowers the conductivity）.The concentration of these imperfections depends on temperature，composition，and the degree of cold work of a metal specimen.In fact， it has been observed experimentally that the total resistivity of a metal is the sum of the contributions from thermal vibrations， impurities， and plastic deformation；that is，the scattering mechanisms act independently of one another.This may be represented in mathematical form as follows:

$$\rho_{\text{total}} = \rho_t + \rho_i + \rho_d \tag{2-4}$$

in which $\rho_t$， $\rho_i$ and $\rho_d$ represent the individual thermal， impurity， and deformation resistivity contributions，respectively.Equation（2-4）is sometimes known as Matthiessen's rule.Electrical and other properties of copper render it the most widely used metallic conductor.Oxygen-free high-conductivity（OFHC）copper，having extremely low oxygen and other impurity contents，is produced for many electrical applications.Aluminum，having a conductivity only about one-half that of copper，is also frequently used as an electrical conductor.Silver has a higher conductivity than either copper or aluminum；however，its use is restricted on the basis of cost.For some applications，such as furnace heating elements，a high electrical resistivity is desirable.The energy loss by electrons that are scattered is dissipated as heat energy.Such materials must have not only a high resistivity，but also a resistance to oxidation at elevated temperatures and，of course，a high melting temperature.Nichrome，a nickel-chromium alloy，is commonly employed in heating elements.

## 4.Piezoelectric Materials

An unusual property exhibited by a few ceramic materials is piezoelectricity，or，literally，pressure electricity:polarization is induced and an electric field is established across a specimen by the application of external forces.Reversing the sign of an external force（i.e.，from tension to compression）reverses the direction of the field.Piezoelectric materials are utilized in transducers，which are devices that convert electrical energy into mechanical strains，or vice versa.Some other familiar applications that employ piezoelectrics include phonograph cartridges，microphones，speakers，audible alarms，and ultrasonic imaging.Piezoelectric materials include titanates of barium and lead，lead zirconate ammonium dihydrogen phosphate and quartz.This property is characteristic of materials having complicated crystal structures with a low degree of symmetry.The piezoelectric behavior of a polycrystalline specimen may be improved by heating above its Curie temperature and then cooling to room temperature in a strong electric field.

## Words and Phrase

divalent *adj*.化合物二价的

magnetic domain 磁畴

dielectric *n*.电解质，绝缘体；

*adj*.非传导性的

polymer *n*.聚合体，多聚体

receptacles *n*.插座，容器，花托

titanate *n*.钛酸盐

barium *n*.钡

piezoelectric *adj*.压电的

tension *n*.张力

compression *n*.压缩

strain *n*.应力

zirconate *n*.锆酸盐

ammonium *n*.铵

dihydrogen *adj*.二氢的

phosphate *n*.磷酸盐

Curie temperature 居里温度

crystalline *adj*.结晶的

reciprocal *adj*.互惠的，倒数的；*n*.倒数

furnace *n*.火炉，熔炉

scatter *v*.撒播，散开，四散，使分散，驱散

nichrome *n*.镍铬合金

chromium *n*.铬

## Notes

1.For this reason the relative area within the hysteresis loop must be small；it is characteristically thin and narrow.

因此，磁滞回线内的相对面积必须很小，其特点是狭而窄。

2.In terms of hysteresis behavior，a hard magnetic material has a high remanence，coercivity，and saturation flux density，as well as a low initial permeability，and high hysteresis energy losses.

在磁滞特性方面，永磁材料具有高的剩磁、矫顽力和饱和磁通密度，以及低的初始磁导率和高的磁滞能耗。

3.These materials also exhibit a high degree of dimensional stability and mechanical strength.Typical applications include power line and electrical insulation，switch bases，and light receptacles.

这些材料还表现出高度的形稳性和机械强度，典型应用包括电源线和电气绝缘、开关底座和灯插座。

4.Such materials must have not only a high resistivity，but also a resistance to oxidation at elevated temperatures and，of course，a high melting temperature.

这种材料不仅具有高电阻率，而且在高温下也具有抗氧化性，当然也具有高熔点温度。

## Exercises

### 1.Answer the following questions according to the text

（1）What is determined only by the composition of the material？

（2）What are the main characteristics of the hard magnetic material？

（3）Silver has a lower conductivity than either copper or aluminum.Right or wrong？

（4）What is the function of Piezoelectric materials when they are utilized in transducers？

### 2.Translate the following sentences into Chinese according to the text

（1）Consequently，a soft magnetic material must have a high initial permeability and a low coercivity.

（2）Again，hysteresis behavior is related to the ease with which the magnetic domain boundaries move；by impeding domain wall motion，the coercivity and susceptibility are enhanced.

（3）The concentration of these imperfections depends on temperature，composition，and the degree of cold work of a metal specimen.

（4）The piezoelectric behavior of a polycrystalline specimen may be improved by heating above its curie temperature and then cooling to room temperature in a strong electric field.

### 3.Translate the following paragraph into Chinese

Giant Magnetostrictive Material（GMM）is a new functional material which has bi-directional transduction effects，it can be made into Giant Magnetostrictive Actuator（GMA）by using the direct effect，also can be made into sensors by using the converse effect.Compared with piezoelectric actuator，GMA owns the advantages of larger load capacity，rapider response，higher reliability and lower voltage drive，etc.It has a promising prospect in precision positioning，active noise and vibration control and fluid control（pump，valve）fields.

# Unit 1　Emerging System Applications and Technological Trends in Power Electronics

## Text A　Emerging System Applications of Power Electronics

Power electronics （ PE ） is going through an exciting time， with increasingly widespread applications.PE fits a typical definition of high technology， since it draws expertise from three broad technical fields of electrical engineering:circuits and automatic control，magnetic and electric machines， and semiconductor devices and integrated circuits（IC）.Today，PE has outgrown its classic definition.It has evolved to include just about all aspects of electrical and electronic engineering.It includes analog and digital circuits，converter circuits，magnetic and electric machines，linear and nonlinear control，energy generation and storage，system engineering and integration，radiofrequency（RF）circuits，antennas， ICs and monolithic passives，and power semiconductors and ICs.This text summarizes six emerging system applications of PE， which include green energy system integrations；microgrids；all things grid connected；transportation electrification；smart homes， smart buildings， and smart cities；and energy harvesting.

### 1.Green Energy System Integration

One of the most significant system applications of PE is the integration of green energy systems.Solar and wind energy systems are maturing fast to penetrate deeply into the energy market to provide electricity for average consumers.The technologies for individual systems are mature，and recent trends are toward the integration of solar farms with wind farms， particularly with offshore wind farms.To transmit the power from offshore to onshore， new highvoltage direct current（HVDC）transmission technology has been developed and deployed around the world.The new technology leverages the recent progress in PE，such as HV insulated gate bipolar transistors（IGBT）and multilevel voltage-source converter（VSC）circuit topology.

### 2.Microgrids

A microgrid is an autonomous electric power subsystem.It can be DC，AC，or a hybrid of DC and AC.A structured microgrid is integrated with loads，energy sources，storage devices，sensors，and data buses，and is an autonomous subsystem.DC microgrids are particularly advantageous for practice due to their many benefits， which include higher efficiency， more robust system operation， no impedance matching issues， no synchronization， simple waveforms， and a large body of knowledge in PE and systems being directly leveraged.

### 3.All Things Grid Connected

"All things grid connected"is a phrase used to describe the key hardware types that an electronictized grid may encompass.They are grid-tied inverters that connect renewable energy resources to the grid， digitally controlled solid-state power transformers，smart fuses and solid-state circuit breakers，flexible AC transmission systems，static variable reactive power compensators，multilevel power converters for HVDC， bidirectional multiport power and control units for the integration of multiple resources， and energy storage，to name a few.In other words，PE is playing a crucial role in grid modernization.It can be said that in the competition for the dominance of smart grids，those who possess better PE technologies will have the upper hand， as was illustrated recently by the success of new HVDC technology with VSC，together with the breakthrough in HVDC circuit breakers.

## 4.Transportation Electrification

Another important area that is dominant in emerging system applications is transportation electrification.PE is the essential driving force of electrification， at least at the early stages.In 2014， the IEEE Transportation Electrification Initiative （TEI） was transitioned into the IEEE Transportation Electrification Community（TEC）， with Power Electronics Society（PELS）providing its administrative home.The community covers all aspects of transportation electrification， such as electric vehicles， charging stations， grid-connected chargers， more electric airplanes， more electric ships， off-road vehicles，trains and traction，connected vehicles，self-driving cars，and intelligent highways.

## 5.Smart Homes，Smart Buildings，and Smart Cities

A closely related system application， but at a higher level to transportation electrification and sustainable energies，is the initiative for smart homes，smart buildings，and smart cities.Smart homes are a natural growth from the smart grid， since the power will be from the smart grid， together with local renewable generation.The scope of smart homes is broad and ever evolving，but it generally includes the following features:smart appliances， smart lighting， personalized energy， mobility， mobile communication， smart security， and remote personal care.To carry out smart control and monitoring， a smart （remote） sensor network is needed.For smart buildings， the key features include integration of renewable energies， high-voltage ac or medium-voltage DC control and distribution， lighting control， access control，security，and power and cooling control.

The idea of smart cities is very recent but nevertheless can be considered a natural extension of smart homes and smart buildings.The IEEE has a workforce called the IEEE Smart Cities Initiative， of which PELS is a member.The scope of a smart city is even more broad and ever evolving，but its key features include smart water， smart energy， smart buildings， smart waste， smart public services， smart security， and smart mobility.PE will play an important role in providing the backbone for future energy conversion and control intelligence for future smart cities.

## 6.Energy Harvesting

Energy harvesting is gaining much wider recognition lately because of the proliferation of mobile technologies and low-power microelectronics.There are many different forms of energy harvesting.Most prevalent are RF，vibration，thermal，solar，human bodies，and structural energy harvesting.An energy-harvesting （mini） system typically includes four parts:energy-conversion device（s）， harvesting， power conditioning，and storage.All these fields are closely related to PE.

## Words and Phases

power electronics　电力电子

semiconductor device　半导体器件

integrated circuit（IC）　集成电路（IC）

radio frequency（RF）射频（RF）

green energy绿色能源

microgrid　*n*.微电网

transportation electrification　交通电气化

smart　*adj*.智能的

energy harvesting　能量收集

solar/wind farm　太阳能发电厂、风力发电厂

highvoltage direct current（HVDC）transmission高压直流（HVDC）输电

insulated gate bipolar transistors（IGBT）　绝缘栅双极型晶体管（IGBT）

voltage-source converter（VSC）电压源型变流器（VSC）

autonomous　*adj*.自治的

robust　*adj*.鲁棒

synchronization　*n*.同步

solid-state power transformer　固态变压器

solid-state circuit breaker　固态断路器

flexible AC transmission　柔性交流输电

multilevel power converter　多电平功率变换器

electric vehicle　电动汽车

self-driving car　自动驾驶汽车

sensor　*n*.传感器

distribution　*n*.分配

low-power microelectronic低功耗微电子

vibration　*n*.振动

thermal　*adj*.热的

## Notes

1.IEEE

电气和电子工程师协会（IEEE，全称是Institute of Electrical and Electronics Engineers）是一个美国的电子技术与信息科学工程师的协会，是目前世界上最大的非营利性专业技术学会，其会员人数超过40万人，遍布160多个国家。IEEE致力于电气、电子、计算机工程和与科学有关的领域的开发和研究，在太空、计算机、电信、生物医学、电力及消费性电子产品等领域已制定了900多个行业标准，现已发展成为具有较大影响力的国际学术组织。

2.Today，PE has outgrown its classic definition.It has evolved to include just about all aspects of electrical and electronic engineering.It includes analog and digital circuits，converter circuits，magnetic and electric machines，linear and nonlinear control，energy generation and storage，system engineering and integration，radiofrequency（RF）circuits，antennas，ICs and monolithic passives，and power semiconductors and ICs.

现今，PE已经超越了它的经典定义。它已经发展到包括电气和电子工程的所有方面，包括模拟和数字电路，变换器电路，磁和电机，线性和非线性控制，能量产生和存储，系统工程和集成，射频电路，天线，IC和单片无源器件，以及功率半导体和IC。

## Exercises

## 1.Answer the following questions according to the text

（1）Why does PE fit a typical definition of high technology?

（2）What are the characteristics of DC microgrids?

（3）Please list some aspects of transportation electrification.

（4）What are the key features of the smart city?

（5）What are the forms of energy harvesting?

## 2.Translate the following sentences into Chinese according to the text

（1）Solar and wind energy systems are maturing fast to penetrate deeply into the energy market to provide electricity for average consumers.

（2）DC microgrids are particularly advantageous for practice due to their many benefits，which include higher efficiency，more robust system operation，no impedance matching issues，no synchronization，simple waveforms，and a large body of knowledge in PE and systems being directly leveraged.

（3）It can be said that in the competition for the dominance of smart grids，those who possess better PE technologies will have the upper hand，as was illustrated recently by the success of new HVDC technology with VSC，together with the breakthrough in HVDC circuit breakers.

（4）PE will play an important role in providing the backbone for future energy conversion and control intelligence for future smart cities.

## 3.Translate the following paragraph into Chinese

Energy harvesting is gaining much wider recognition lately because of the proliferation of mobile technologies and low-power microelectronics.There are many different forms of energy harvesting.Most prevalent are RF，vibration，thermal，solar，human bodies，and structural energy harvesting.An energy-harvesting（mini）system typically includes four parts:energy-conversion device（s），harvesting，power conditioning，and storage.All these fields are closely related to PE.

# Unit 2　Overview of Power Semiconductor Devices

## Text A　Silicon Carbide Power Semiconductor Devices

Power semiconductor devices are attracting increasing attention as key components in a variety of power electronic systems.The major applications of power devices include power supplies， motor controls，renewable energy，transportation，telecommunications，heating，robotics，and electric utility transmission/distribution.The utilization of semiconductor power devices in these systems can enable significant energy savings，increased conservation of fossil fuels，and reduced environmental pollution.

Power electronics has gained renewed attention in the past decade due to the emergence of several new markets， including converters for photovoltaic and fuel cells， converters and inverters for electric vehicles（EVs）and hybrid-electric vehicles（HEVs），and controls for smart electric utility distribution grids.Currently， semiconductor power devices are one of the key enablers for global energy savings and electric power management in the future.

Silicon power devices have improved significantly over the past several decades，but these devices are now approaching performance limits imposed by the fundamental material properties of silicon，and further progress can only be made by migrating to more robust semiconductors.Silicon carbide（SiC）is a wide-bandgap semiconductor with superior physical and electrical properties that can serve as the basis for the high-voltage，low-loss power electronics of the future.

SiC is a IV-IV compound semiconductor with a bandgap of 2.3～3.3 e V（depending on the crystal structure， or polytype）.It exhibits about 10 times higher breakdown electric field strength and 3 times higher thermal conductivity than silicon， making it especially attractive for high-power and high-temperature devices.For example， the on-state resistance of SiC power devices is orders-of-magnitude lower than that of silicon devices at a given blocking voltage，leading to much higher efficiency in electric power conversion.The wide bandgap and high thermal stability make it possible to operate certain types of SiC devices at junction temperatures of 300 ℃ or higher for indefinite periods without measurable degradation.Among wide-bandgap semiconductors，SiCis exceptional because it can be easily doped either p-type or n-type over a wide range， more than five orders-of-magnitude.In addition， SiC is the only compound semiconductor whose native oxide is $SiO_2$，the same insulator as silicon.This makes it possible to fabricate the entire family of MOS-based（metal-oxidesemiconductor）electronic devices in SiC.

Since the 1980s， sustained efforts have been directed toward developing SiC material and device technology.Based on a number of breakthroughs in the 1980s and 1990s， SiC Schottky barrier diodes（SBDs）were released as commercial products in 2001.The market for SiC SBDs has grown rapidly over the last several years.SBDs are employed in a variety of power systems，including switch-mode power supplies， photovoltaic converters， air conditioners， and motor controls for elevators and subways.Commercial production of SiC power switching devices，primarily JFETs（junction field-effect transistors）and MOSFETs（metal-oxide-semiconductor field-effect transistors），began in 2006-2010.These devices are well accepted by the markets and many industries are now taking advantage of the benefits of SiC power switches.As an example，the volume and weight of a power supply or inverter can be reduced by a factor of 4～10，depending on the extent to which SiC components are employed.In addition to the size and weight reduction， there is also a substantial reduction in power dissipation， leading to improved efficiency in electric power conversion systems due to the use of SiC components.

### Words and Phases

Power semiconductor device　功率半导体器件

power supply　电源

fuel cell　燃料电池

wide-bandgap　*n.*宽禁带

compound　*adj.*复合的

polytype　*n.*多型体

on-state voltage　导通电压

blocking voltage　阻断电压

junction temperature　结温

native oxide　本征氧化层

Schottky barrier diode　肖特基势垒二极管

photovoltaic converter　光伏变换器

metal-oxide-semiconductor field-effect transistor　金属氧化物半导体场效应晶体管

## Note

1.Silicon power devices have improved significantly over the past several decades，but these devices are now approaching performance limits imposed by the fundamental material properties of silicon， and further progress can only be made by migrating to more robust semiconductors.

过去几十年，硅基功率器件的性能获得大幅提升，但受硅材料特性的制约，硅基功率器件性能已发挥到极限，为了进一步提升功率器件的性能，需要采用性能更加优异的半导体材料。

2.It exhibits about 10 times higher breakdown electric field strength and 3 times higher thermal conductivity than silicon，making it especially attractive for high-power and hightemperature devices.

与硅相比，它（碳化硅）具有高于10倍的击穿电场强度和高于三倍的热导率，这使得它（碳化硅）对制备高功率、高温器件更具吸引力。

3.In addition to the size and weight reduction， there is also a substantial reduction in power dissipation， leading to improved efficiency in electric power conversion systems due to the use of SiC components.

除了体积和重量的减少，碳化硅器件的使用还可使功率损耗大幅降低，从而提升电能变换系统的效率。

## Exercises

## 1.Answer the following questions according to the text

（1）What is the most common application of power semiconductor devices?

（2）What advanced properties does SiC possess compared with Si?

（3）Why can SiC devices operate at higher junction temperature?

（4）What is the mechanism behind the reduction of the size and weight of a power converter using SiC power devices?

## 2.Translate the following sentences intoChinese according to the text

（1）The major applications of power devices include power supplies，motor controls，renewable energy，transportation，telecommunications，heating，robotics，and electric utility transmission/distribution.

（2）The utilization of semiconductor power devices in these systems can enable significant energy savings，increased conservation of fossil fuels，and reduced environmental pollution.

（3）Silicon carbide（SiC）is a wide-bandgap semiconductor with superior physical and electrical properties that can serve as the basis for the high-voltage，low-loss power electronics of the future.

（4）As an example，the volume and weight of a power supply or inverter can be reduced by a factor of 4～10，depending on the extent to which SiC components are employed.

## 3.Translate the following intoChinese

The wide bandgap and high thermal stability make it possible to operate certain types of SiC devices at junction temperatures of 300 ℃ or higher for indefinite periods without measurable degradation.Among wide-bandgap semiconductors，SiC is exceptional because it can be easily doped either p-type or n-type over a wide range，more than five orders-of-magnitude.In addition，SiC is the only compound semiconductor whose native oxide is $SiO_2$，the same insulator as silicon.This makes it possible to fabricate the entire family of MOS-based（metal-oxide-semiconductor）electronic devices in SiC.

Since the 1980s，sustained efforts have been directed toward developing SiC material and device technology.Based on a number of breakthroughs in the 1980s and 1990s，SiC Schottky barrier diodes（SBDs）were released as commercial products in 2001.The market for SiC SBDs has grown rapidly over the last several years.SBDs are employed in a variety of power systems，including switch-mode power supplies，photovoltaic converters，air conditioners，and motor controls for elevators and subways.Commercial production of SiC power switching devices，primarily JFETs（junction field-effect transistors）and MOSFETs（metal-oxide-semiconductor field-effect transistors），began in 2006-2010.These devices are well accepted by the markets and many industries are now taking advantage of the benefits of SiC power switches.

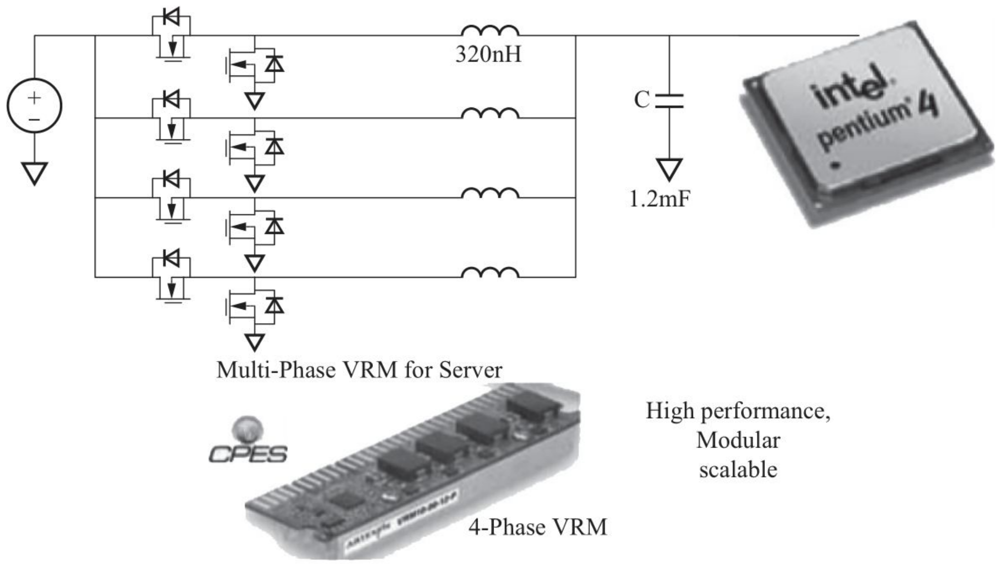
# Unit 4　DC-DC Converter

## Text A　Trend for Distributed Power Systems

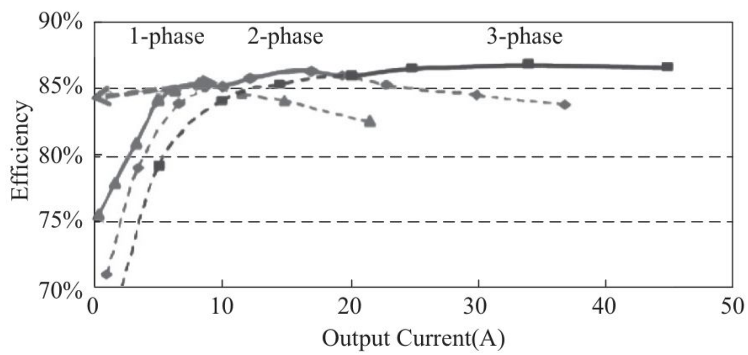
Power electronics products， to date， are essentially custom-designed.With a long design cycle， today's power electronics equipment is designed and manufactured using non-standard parts.Thus， manufacturing processes are labor-intensive， resulting in high cost and poor reliability.This practice has significantly weakened the U.S.power electronics industry in recent years.In the 1980s，power electronics was considered to be a core enabling technology for all of the major corporations in the U.S.In the 1990s， the major corporations adopted an outsourcing strategy and spun off their power electronics divisions.What had been a captive market was transformed into a merchant market.Fewer resources were available to devote to technical advancements in power electronics.Therefore，innovative solutions were scarce，and products became commoditized and cost-driven.The result was the increased mass migration of manufacturing to countries with low labor costs.The problem is further compounded by the more recent trend of outsourcing engineering to India and Asian countries，especially China.Today，most of the industry is focused on the bottom line and spends little on R and D，and of this most is spent on development rather than on research.

With ever-increasing current consumption（＞100 A）and clock frequency（＞GHz），today's microprocessors are operating at very low voltages（1 V or less）and continuously switching between the sleep-mode and wake-up mode at frequencies of up to several MHz to conserve energy.Lee has proposed a multiphase voltage regulator（VR）module for new generations of Intel Pentium microprocessors（Fig.3.5）.

This multiphase VR technology is simple and is easily scalable to meet ever-increasing current consumption，clock rates，and stringent voltage regulation requirements.With the ability to adjust phases according to the load demand，it demonstrates superior efficiency at light load，at which microprocessors operate most of the time.Today，every PC and server microprocessor in the world is powered with this VR.These technologies have been further extended to high-performance graphical processors，server chipset and memory devices，networks，telecommunications，and all forms of mobile electronics.For example，in telecommunication applications，48 V has been adopted as the low-voltage bus for all switch boards.A number of isolated DC/DC converters，often referred to as brick converters，are employed for conversion from 48 V to low-voltage outputs in each switch board.The brick converters demand high power density and high efficiency，and thus are very competitive products and laden with IPs.With the advent of multiphase VRs，these brick products have been quickly converted into a two-stage solution，with a simple DC/DC transformer（DCX）followed by a multiphase VR，as shown in Fig.3.6，with demonstrated improvement in efficiency，power density，and even cost.With the widespread use of distributed point-of-Load，the power architecture has undergone significant changes，from centralized customized power supplies to more distributed power supplies（DPS），as shown in Fig.3.7.In this DPS architecture，the modular approach has been adopted for even the front-end power conversion.The DPS architecture offers many benefits，such as modularity，scalability，fault tolerance，improved reliability，serviceability，redundancy，and a reduced time to market.




(a)




(b)

Fig.3.5　A multiphase voltage regulatot（VR）module for new generation of Intel Pentium microprocessors

（a）Multiphase VR for the new generation of microprocessors；（b）Efficiency improvement at light load using phase shedding and burst-mode operation

## Words and Phases

multiphase　*n*.多相

centralized　*adj*.集中式的

custom-designed　*adj*.定制的

outsourcing　*adj*.外包的

chipset　*n*.芯片集

switch boards　开关板

architecture　*n*.结构

modularity　*n*.模块性

fault tolerance　容错性

redundancy　*n*.冗余性

## Notes

1.With ever-increasing current consumption（＞100 A）and clock frequency（＞GHz），today's microprocessors are operating at very low voltages（1 V or less）and continuously switching between the sleep-mode and wake-up mode at frequencies of up to several MHz to conserve energy.

随着电流消耗（大于100A）和时钟频率（大于GHz）的不断增加，今天的微处理器在非常低的电压（1V或更小）下工作，并在睡眠模式和唤醒模式之间持续切换，频率高达几兆赫，以节省能源。

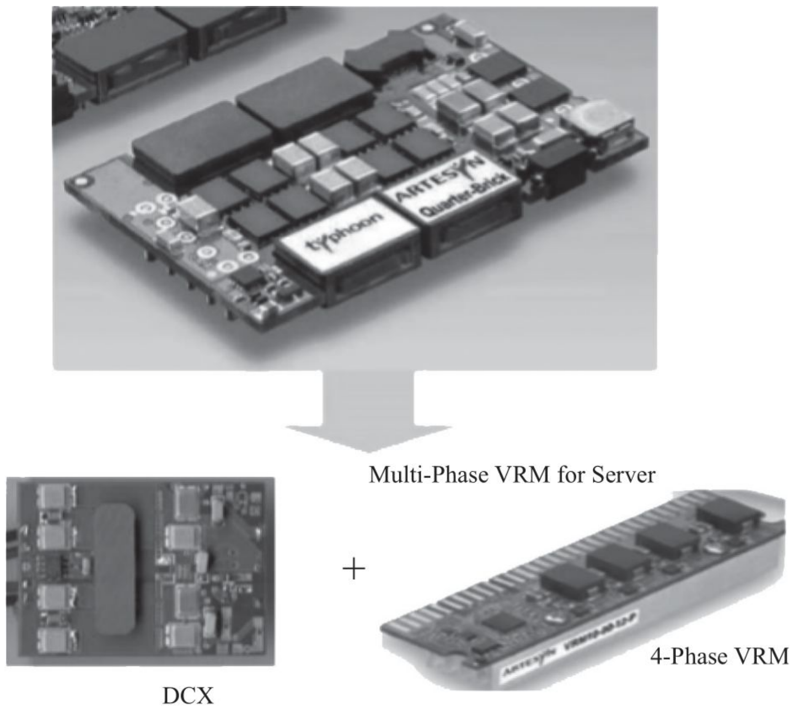



Fig.3.6　DC/DC brick converter is replaced with DCX and POL VR

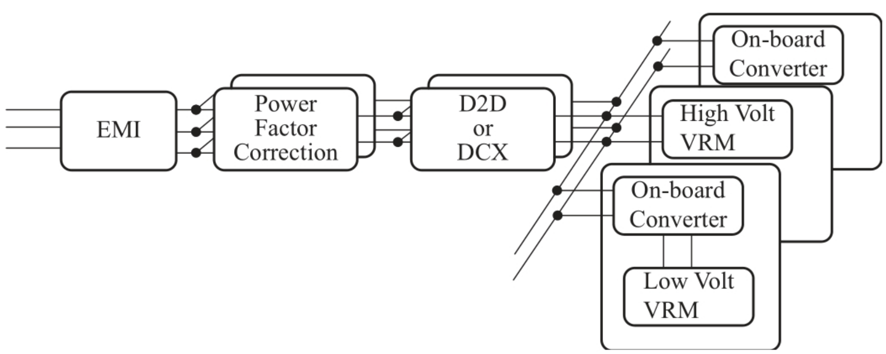



Fig.3.7　Distributed power architecture

2.With the widespread use of distributed point-of-Load，the power architecture has undergone significant changes，from centralized customized power supplies to more distributed power supplies（DPS）.

随着分布式负载点的广泛使用，电源结构发生了重大变化，从集中式定制电源到更加分布式的电源（DPS）。

## Exercises

### 1.Answer the following questions according to the text

（1）Why is the U.S.power and electronics industry weakened？

（2）What are the reasons why products in most industries are becoming commercialized and cost-driven today？

（3）What are the characteristics of multi-phase virtual reality technology？

（4）What major changes have taken place in the power supply structure？

（5）What are the advantages of DPS architecture？

### 2.Translate the following sentences into Chinese according to the text

（1）In the 1990s，the major corporations adopted an outsourcing strategy and spun off their power electronics divisions.

（2）Today，most of the industry is focused on the bottom line and spends little on R and D，and of this most is spent on development rather than on research.

（3）This multiphase VR technology is simple and is easily scalable to meet everincreasing current consumption，clock rates，and stringent voltage regulation requirements.

（4）These technologies have been further extended to high-performance graphical processors，server chipset and memory devices，networks，telecommunications，and all forms of mobile electronics.

（5）The DPS architecture offers many benefits，such as modularity，scalability，fault tolerance，improved reliability，serviceability，redundancy，and a reduced time to market.

### 3.Translate the following into Chinese

With the continuous progress of semiconductor technology and the development of integrated technology，the characteristic size of microprocessor chips is getting smaller and smaller.Because of the process requirements，and also in order to obtain the lowest power consumption and the highest efficiency，the power supply voltage of bit processors tends to be low-voltage.With the decrease of the working voltage of bit processors，their working voltage becomes smaller and smaller.The requirement of stability becomes higher.Voltage fluctuation of 50 m V may lead to wrong operation of the circuit.At the same time，the stronger the function of microprocessor，the larger the circuit size and internal function circuit，and the larger the driving current provided by the power supply.At the same time，microprocessor needs to work according to different work in order to reduce power consumption.In order to maintain high voltage stability under the condition of low voltage，high current and high current change rate，the power supply module of microprocessor is required to have excellent dynamic performance.

# Unit 5　DC-AC Converter

## Text A　Transformerless Inverters

In photovoltaic（PV）applications，a transformer is often used to provide galvanic isolation and voltage ratio transformations between input and output.However，these conventional iron and copper-based transformers increase the weight/size and cost of the inverter whilst reducing the efficiency and power density.It is therefore desirable to avoid using transformers in the inverter.However，additional care must be taken to avoid safety hazards such as ground fault currents and leakage currents，e.g.via the parasitic capacitor between the PV panel and ground.Consequently，the grid connected transformerless PV inverters must comply with strict safety standards such as interconnection and interoperability requirements standard for relevant interfaces between interconnected distributed energy and power system IEEE 1547.1，standard requirements for automatic disconnection device between generator and public low voltage grid VDE0126-1-1，safety standard for household and similar electrical appliances EN 50106，standard for characteristics of the utility interface in PV systems IEC61727，and the latest standards for the installation of solar panels AS/NZS 5033.

Various transformerless inverters have been proposed recently to eliminate the leakage current using different techniques such as decoupling the DC from the AC side and/or clamping the common mode（CM）voltage during the freewheeling period，or using common ground configurations.The permutations and combinations of various decoupling techniques with integrated voltage buck-boost for maximum uepower point tracking（MPPT）allow numerous new topologies and configurations which are often confusing and difficult to follow when seeking to select the right topology.Therefore，to present a clear picture on the development of transformerless inverters for the next generation grid-connected PV systems，this paper aims to comprehensively review and classify various transformerless inverters with detailed analytical comparisons.To reinforce the findings and comparisons as well as to give more insight on the CM characteristics and leakage current，computer simulations of major transformerless inverter topologies have been performed in PLECS software.Moreover，the cost and size are analysed properly and summarized in a table.Finally，efficiency and thermal analysis are provided with a general summary as well as a technology roadmap.

Solar photovoltaic（PV）is one of the cleanest，readily and widely available energy sources among all renewable energies.With technological advancements in material and manufacturing techniques，the cost of the PV system is continuously reduced，making it the cheapest energy source for massive deployment in the future.Many countries（USA，Germany，China，Japan，Australia，France，Italy，Spain，etc.）have already begun to reap the benefits through their increased adoption and integration of this system in the utility grid.According to the 2017 annual report of the International Energy Agency-Photovoltaic Power Systems Program（IEA-PVPS），the global installed PV capacity reached a 100 GW milestone in 2012，and a 200 GW level in 2015.By the end of 2017，the total installed PV capacity was estimated to be roughly 410 GW，while 24 IEA-PVPS countries reached 264 GW.Fig.3.8 shows the cumulative installed PV capacity of the top IEA-PVPScountries from 2012 to 2017.From this figure，it is evident that the PV industry is facing rapid growth with five leading countries representing 90.1%of all PVinstallations in 2017.Among them，China，USA，and Japan experienced the largest installed PV installation capacity increment in recent years.

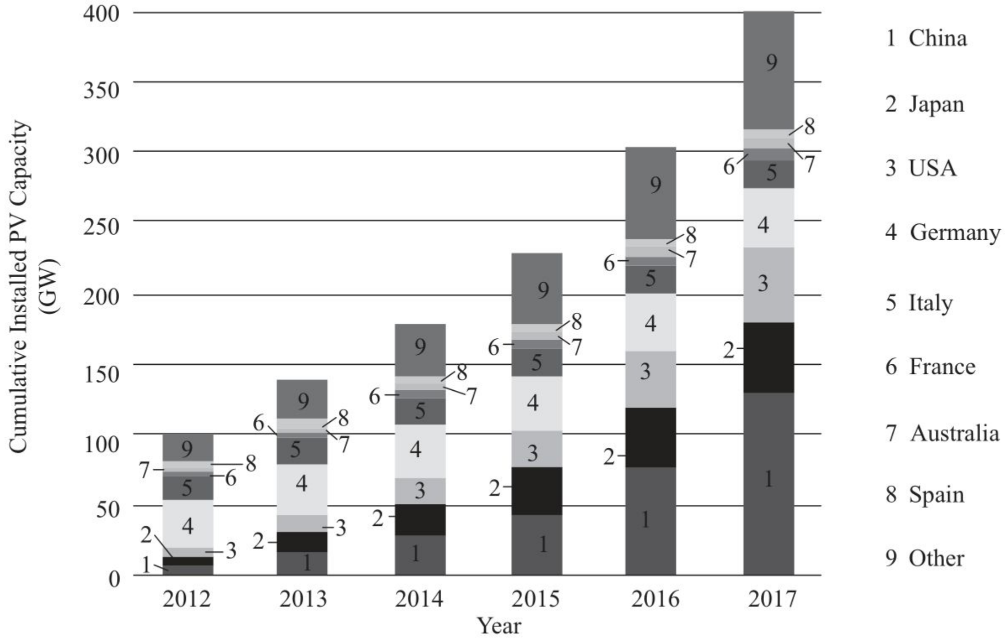



Fig.3.8 Cumulative PV installations for the top IEA-PVPS countries from 2012 to 2017

Among all PV installations, the percentage of off-grid PV systems is very low.The grid-connected PV systems need power inverters as interfaces between the PV panel and the grid, and these are generally categorised as galvanic isolated inverters and non-isolated inverters.In the isolated type, usually a high-frequency DC side transformer or a lowfrequency AC side transformer is used to achieve galvanic isolation and this serves to enhance the overall system safety.Due to their lower cost, size/weight, and higher efficiency, transformerless inverters have generated a high degree of interest in terms of residential market with low to medium power capacity.

Fig.3.9 illustrates a general layout for a single-phase transformerless inverter for smallscale PV systems.As can be seen, without a galvanic isolation, a direct ground-current path may form between the PV panel and the grid.Due to the presence of large stray capacitance ($C_{PV}$) between the PV and grid grounds, the varying voltage (also known as common-mode (CM) voltage) can excite the resonant circuit formed by the parasitic capacitor and inverter filter inductor and this produces a high CM ground current $i_{dn}$.This capacitive $i_{dn}$ comprises line low-frequency and switching high-frequency components which inject harmonics into the grid current, increase the system losses, impair the electromagnetic compatibility, and can cause safety problems such as electric shock.

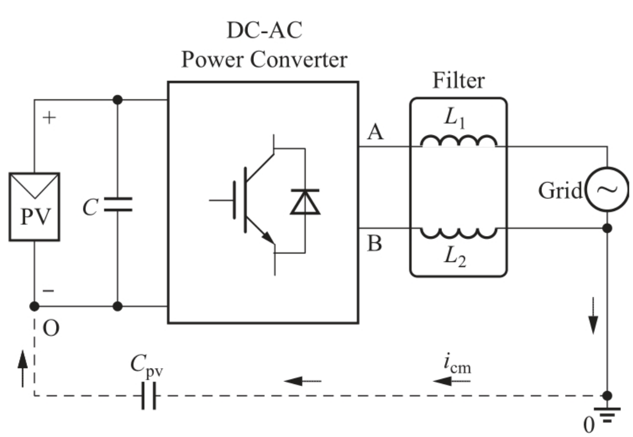


Fig.3.9　The general layout of a single-phase transformerless inverter using an L-filter

In order to understand the grid-connected PV systems to satisfy various grid codes and their safety standards， numerous inverter related issues have been thoroughly investigated.So far， many transformerless inverter topologies have been presented with the aim of eliminating the leakage current.To achieve this，various decoupling techniques have been adopted，such as，decoupling the DC from the AC side and/or clamping the common mode voltage（CMV）during the freewheeling period，or using common ground configurations.The combinations of these decoupling techniques with integrated MPPT circuits form an immense number of topologies and configurations which are often confusing and difficult to follow.Therefore，to present a clear picture on the development of the transformerless inverter for the next generation grid-connected PV systems，this paper aims to review and classify various transformerless inverters.Further，it aims to provide an analytical overview and analysis of well-known single-phase transformerless inverter topologies as well as a comparison of the transformerless inverters based on a loss and efficiency analysis through the means of detailed calculations.This categorisation and analysis can help researchers to understand the advantages and disadvantages of various transformerless inverter topologies in terms of their CMV and leakage current behaviour.

## Words and Phases

galvanic isolation　电气隔离

ground fault currents　接地故障电流

leakage currents　漏电流

parasitic capacitor　寄生电容

freewheeling period　续流周期

cumulative　*adj.*累积的，渐增的

stray capacitance　杂散电容

resonant circuit　谐振电路

transformerless inverters　非隔离逆变器

maximum power point tracking（MPPT）　最大功率点跟踪

## Notes

1.It is therefore desirable to avoid using transformers in the inverter.However，additional care must be taken to avoid safety hazards such as ground fault currents and leakage currents，e.g.via the parasitic capacitor between the PV panel and ground.

因此，最好避免在逆变器中使用变压器。但是，必须采取额外的注意措施，以避免诸如接地故障电流和漏电流等危险，例如光伏电池板和接地之间的寄生电容引入。

2.Various transformerless inverters have been proposed recently to eliminate the leakage current using different techniques such as decoupling the DC from the AC side and/or clamping the common mode（CM）voltage during the freewheeling period，or using common ground configurations.

为了消除漏电流，最近提出了各种无变压器逆变器，如将直流与交流侧在续流阶段解耦或钳位共模电压，或采用共同接地方式。

## Exercises

## 1.Answer the following questions according to the text

（1）Why is it best to avoid using transformers in inverters？

（2）There are several forms of transformerless inverter？

（3）Why are more and more countries paying attention to solar photovoltaic？

（4）What happens without current isolation in the overall layout of single-phase transformer-free inverters for small PV systems？

（5）What is the effect of voltage change caused by stray capacitance？

## 2.Translate the following sentences into Chinese according to the text

（1）The permutations and combinations of various decoupling techniques with integrated voltage buck-boost for maximum power point tracking （MPPT） allow numerous new topologies and configurations which are often confusing and difficult to follow when seeking to select the right topology.

（2）With technological advancements in material and manufacturing techniques，the cost of the PV system is continuously reduced，making it the cheapest energy source for massive deployment in the future.

（3）The grid-connected PV systems need power inverters as interfaces between the PV panel and the grid，and these are generally categorised as galvanic isolated inverters and nonisolated inverters.

（4）In the isolated type，usually a high-frequency DC side transformer or a lowfrequency AC side transformer is used to achieve galvanic isolation and this serves to enhance the overall system safety.

（5）Due to the presence of large stray capacitance（$C_{\mathrm{PV}}$）between the PV and grid grounds，the varying voltage（also known as common-mode（CM）voltage）can excite the resonant circuit formed by the parasitic capacitor and inverter filter inductor and this produces a high CM ground current $i_{\mathrm{dn}}$.

## 3.Translate the following into Chinese

Depending on PV modules configuration，inverters are mainly categorized as central，String，Multi-string and Module inverters.The central inverters are used for three phase large power applications while others String，Multi-string and Module inverters can be three or single phase depending on power rating and type of applications.

Furthermore，single phase PV inverters can be classified on the basis of galvanic isolation such as with-transformer andtransformerless.The detailed classification of grid interactive PV inverter topologies are illustrated in Fig.3.8.

# Unit 3　Electrical Power System

## Text A　Components of Power System

Fig.4.4 shows a minimum power system.The system consists of an energy source，a prime mover，a generator， and a load.The energy source may be coal，gas，or oil burned in a furnace to heat water and generator steam in a boiler.The prime mover may be a steam turbine，a hydraulic turbine or water wheel，or an internal combustion engine.Each one of these prime movers has the ability to convert energy in the form of heat，falling water，or fuel into rotation of a shaft，which in turn will drive the generator.The electrical load on the generator may be lights， motors， heaters， or other devices， alone or in combination.The control system functions to keep speed of the machines substantially constant and the voltage within prescribed limits， even the load may change.The control system may include a main stationed in the power plant who watches a set of meters on the generator-output terminals and makes the necessary adjustments manually.In a modern station，the control system is a servomechanism that senses a generator-output conditions and automatically makes the necessary changes in energy input and field current to hold the electrical output within certain specifications.

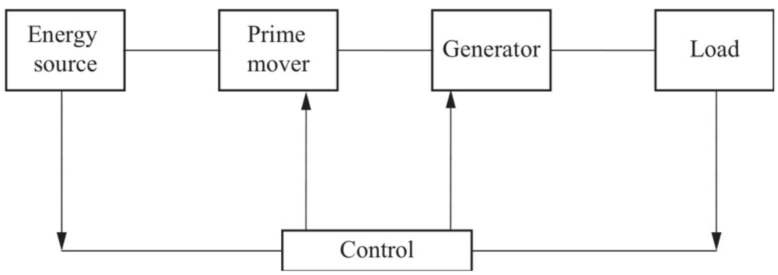



Fig.4.4　Minimum power system

Modern power systems are usually large-scale， geographically distributed， and with hundreds to thousands of generators operating in parallel and synchronously.They may vary in size and structure from one to another，but they all have basic characteristics:

（1）Are comprised of three-phase AC systems operating essentially at constant voltage.Generation and transmission facilities use three-phase equipment.Industrial loads are invariably three-phase；single-phase residential and commercial loads are distributed equally among the phases so as to effectively form a balanced three-phase system.

（2）Use synchronous machines for generation of electricity.Prime movers convert the primary energy（fossil，nuclear，and hydraulic）to mechanical energy that is，in turn，converted to electrical energy by synchronous generators.

（3）Transmit power over significant distances to consumers spread over a wide area.This requires a transmission system comprising subsystems operating at different voltage levels.

Fig.4.5 shows a basic element of a modern power system.Electric power is produced at generating stations（GS）and transmitted to consumers through a complex network of individual components，including transmission lines，transformers，and switching devices.It is common practice to classify the transmission network into the following subsystems:transmission system；subtransmission system；distribution system.

The transmission system interconnects all major generating stations and main load centers in the system.It forms the backbone of the integrated power system and operates at the highest voltage levels.The generator voltages are usually in the range of 11 kV to 35 kV.These are stepped up to the transmission voltage level， and power is transmitted to transmission substations where the voltages are stepped down to the subtransmission level（typically，69 kV to 138 kV）.The generation and transmission subsystems are often referred to as the bulk power system.

The subtransmission system transmits power in small quantities from the transmission substations to the distribution substations.Large industrial customers are commonly supplied directly from the subtransmission system.In some systems， there is no clear demarcation between subtransmission and transmission circuits.As the system expands and higher voltage levels become necessary for transmission， the older transmission lines are often relegated to subtransmission function.

The distribution system represents the final stage in the transfer of power to the individual customers.The primary distribution voltage is typically between 4.0 kV and 34.5 kV.Small industrial customers aresupplied by primary feeders at this voltage level.The secondary distribution feeds supply residential and commercial customers at 120/240 V.

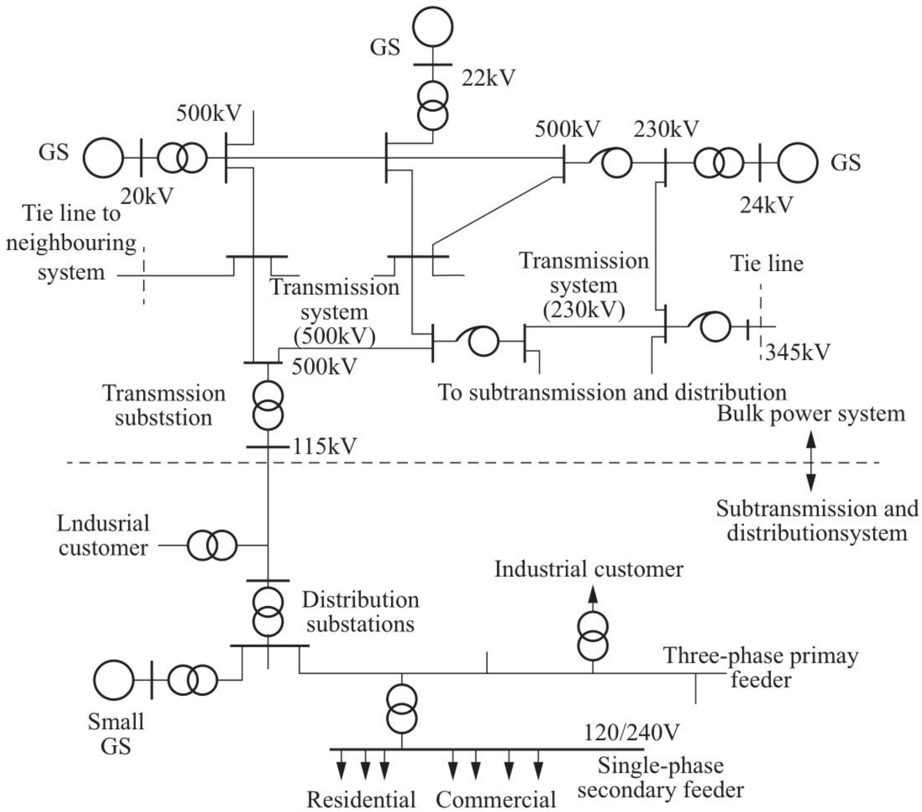



Fig.4.5 Basic elements of a modern power system

Small generating plants located near the load are also connected the subtransmission or distribution system directly.Interconnections to neighboring power systems are usually formed at the transmission system level.The overall system thus consists of multiple generating sources and several layers of transmission networks.This provides a high degree of structural redundancy that enables the system to withstand unusual contingencies without service disruption to the customers.

## Words and Phases

fossil　*n.*化石，石块

prime mover　原动机

steam turbine　汽轮机

hydraulic turbine　水轮机

internal combustion engine　内燃机

servomechanism　*n.*伺服机构

mechanical energy　机械能

bulk power system　大容量电力系统

contingencies　*n.*偶然事故，意外事故

## Notes

1.The control system functions to keep speed of the machines substantially constant and the voltage within prescribed limits，even the load may change.

控制系统的作用是在负载变化的情况下仍能保持电机的基本稳定并将电压控制在规定的范围之内。

2.The control system may include a man stationed in the power plant who watches a set of meters on the generator-output terminals and makes the necessary adjustments manually.

控制系统也可能包括一位派守在电厂的值班员，该值班员观察发电机输出端的一套仪表，并做出一些必要的手动调整。

3.Modern power systems are usually large-scale， geographically distributed， and with hundreds to thousands of generators operating in parallel and synchronously.

现代电力系统通常规模大，地域广，具有成百上千并列同步发行的电机组。

## Exercises

### 1.Answer the following questions according to the text

（1）How many componentsis the power system comprised of?

（2）What is the function of the power system control？

（3）What is the function oftransmission system？

（4）What is the function of subtransmission system？

（5）What are the characteristics of a modern power system in network structure？

### 2.Translate the following sentences into Chinese according to the text

（1）Each one of these prime movers has the ability to convert energy in the form of heat， falling water，or fuel into rotation of a shaft，which in turn will drive the generator.

（2）In a modern station， the control system is a servomechanism that senses a generator-output condition and automatically makes the necessary changes in energy input and field current to hold the electrical output within certain specifications.

（3）Industrial loads are invariably three-phase， single-phase residential and commercial loads are distributed equally among the phases so as to effectively form a balanced threephase system.

（4）Electric power is produced at generating stations（GS）and transmitted to consumers through a complex network of individual components， including transmission lines， transformers， and switching devices.

（5）This provides a high degree of structural redundancy that enables the system to withstand unusual contingencies without service disruption to the customers.

## 3.Translate the following sentences into Chinese

In planning an electric utility system， the physical location of the generating station， transmission lines and substations must be carefully planned to arrive at an acceptable， economic solution.We can sometimes locate a generating station next to the source of energy（such as a coal mine）and use transmission lines to carry the electrical energy to where it is needed.When this is neither practical economical， we have to transport the energy（coal， gas， oil）by ship， train， or pipeline to the generating station.The generating station may，therefore，be near to，or far from，the ultimate user of the electrical en energy.Some of the obstacles prevent transmission lines from following the shortest route.Due to these obstacles， both physical and legal， transmission lines often follow a zigzag path between the generating station and the ultimate user.

# Unit 3 Insulation Testing of Electrical Apparatus

## Text A Measurement of High Voltages

In industrial testing and research laboratories, it is essential to measure the voltages accurately, ensuring perfect safety to the personnel and equipment.Hence a person handling the equipment as well as the metering devices must be protected against overvoltages and also against any induced voltages due to stray coupling.Electromagnetic interference is a serious problem in impulse voltage measurements, and it has to be avoided or minimized.Therefore, even though the principles of measurements may be same, the devices and instruments for measurement of high voltages differ vastly from the low voltage devices.Different devices used for high voltage measurements may be classified as in Tab.5.2.

Tab.5.2 High voltage measurement techniques

| Type of voltage | Method or technique |
|---|---|
| DC voltages | Series resistance microammeter |
| | Resistance potential divider |
| | Generating voltmeters |
| power frequency voltages | Series impedance ammeters |
| | Potential dividers (resistance or capacitance type) |
| | Potential transformers |
| | Electrostatic voltmeters |
| AC high frequency voltages, impulse voltages, and other rapidly changing voltages | Potential dividers with a cathode ray oscillograph (resistive or capacitive dividers) |
| | Peak voltmeters |

### 1.Measurement of High DC Voltages

Measurement of high DC voltages as in low voltage measurements, is generally accomplished by extension of meter range with a large series resistance.The net current in the meter is usually limited to one to ten microamperes for full-scale deflection.For very high voltages (1000 kV or more) problems arise due to large power dissipation, leakage currents and limitation of voltage stress per unit length, change in resistance due to temperature variations, etc.Hence, a resistance potential divider with an electrostatic voltmeter is sometimes better when high precision is needed.But potential dividers also suffer from the disadvantages stated above.Both series resistance meters and potential dividers cause current drain from the source.Generating voltmeters are high impedance devices and do not load the source.They provide complete isolation from the source voltage (high voltage) as they are not directly connected to the high voltage terminal and hence are safer.

### 2.Measurement of High AC Voltages

Measurement of high AC voltages employs conventional methods like series impedance voltmeters, potential dividers, potential transformers, or electrostatic voltmeters.But their designs are different from those of low voltage meters, as the insulation design and source loading are the important criteria.

### 3.Series Impedance Voltmeters

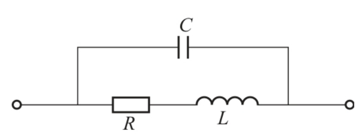



Fig.5.2 Simplified lumped parameter equivalent circuit of a high ohmic resistance R

*L*—Residual inductance; *C*—Residual capacitance

For power frequency AC measurements the series impedance may be a pure resistance or a reactance.Since resistances involve power losses，often a capacitor is preferred as a series reactance.Moreover，for high resistances，the variation of resistance with temperature is a problem， and the residual inductance of the resistance gives rise to an impedance different from its ohmic resistance.High resistance units for high voltages have stray capacitances and hence a unit resistance will have an equivalent circuit as shown in Fig.5.2.At any frequency$\omega$ of the AC voltage，the impedance of the resistance $R$ is

$$Z=\frac{R+\mathrm{j}\omega L}{(1-\omega^2 LC)+\mathrm{j}\omega CR}$$

If$\omega L$ and$\omega C$ are small compared to $R$，

$$Z\approx R\left[1+\mathrm{j}\left(\frac{\omega L}{R}-\omega CR\right)\right]$$

and the total phase angle is

$$\varphi\approx\left(\frac{\omega L}{R}-\omega CR\right)$$

This can be made zero and independent of frequency，if

$$L/C=R^2$$

For extended and large dimensioned resistors， this equivalent circuit is not valid and each elemental resistor has to be approximated with this equivalent circuit.The entire resistor unit then has to be taken as a transmission line equivalent， for calculating the effective resistance.Also， the ground or stray capacitance of each element influences the current flowing in the unit，and the indication of the meter results in an error.The equivalent circuit of a high voltage resistor neglecting inductance and the circuit of compensated series resistor using guard and tuning resistors is shown in Fig.5.3（a）and（b）respectively.Stray capacitance to ground effects［refer Fig.5.3（b）］can be removed by shielding the resistor $R$ by guard resistor $R_s$，which shunts the actual resistor but does not contribute to the current through the instrument.By tuning the resistors $R_a$，the shielding resistor end potentials may be adjusted so that the resulting compensation currents between the shield and the measuring resistors provide a minimum phase angle.

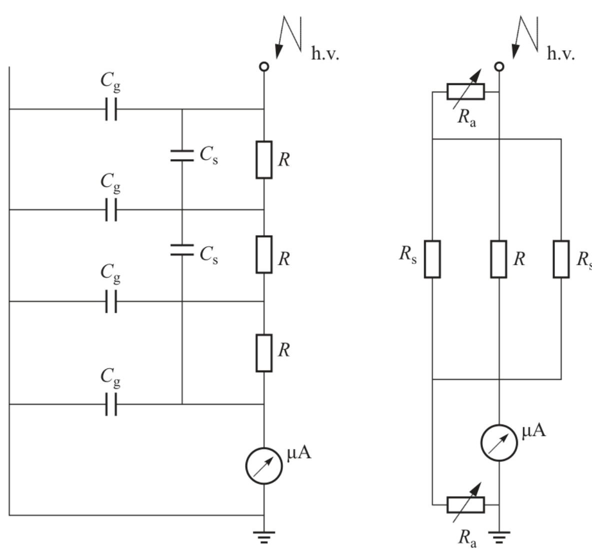



Fig.5.3　Extended series resistance for high AC voltage measurements

（a）Extended series resistance with inductance neglected；（b）Series resistance with guard and tuning resistances
$C_g$—Stray capacitance to ground；$C_s$—Winding capacitance；$R$—Series resistor；$R_s$—Guard resistor；$R_a$—Tuning resistor

## Words and Phases

metering device　测量装置，计量仪表

induced　*adj*.感应的，感生的

stray coupling　杂散耦合，寄生耦合

micro-ammeter　微安计，微安表

potential divider　分压器

series impedance　串联阻抗

ammeter　*n*.电流表，安培计

cathode ray oscillograph　阴极射线示波器

net current　净电流

microampere　*n*.（电流单位）微安培

full-scale deflection　满刻度偏转，最大偏转

power dissipation　功率耗散，功率消耗

precision　*n*.精确，精密度，精度

source　*n*.源，电源

power loss　功率损耗

residual　*adj*.剩余的，残留的

ohmic　*adj*.电阻性的，欧姆性的

lumped parameter　集中参数测量，集总参量

shunt　*v*.分流，*n*.分路器，分流器

## Notes

1.Stray capacitance to ground effects［refer Fig.3.2（b）］can be removed by shielding the resistor $R$ by guard resistor $R_s$，which shunts the actual resistor but does not contribute to the current through the instrument.

如图3.2（b）所示，对地的寄生电容效应通过保护电阻$R_s$屏蔽电阻$R$而消除。$R_s$只分流实际电阻而不通过仪表贡献电流。

2.By tuning the resistors $R_a$，the shielding resistor end potentials may be adjusted so that the resulting compensation currents between the shield and the measuring resistors provide a minimum phase angle.

通过调节电阻$R_a$，保护电阻两端电势可调，使在保护电阻和被测电阻间相应变化的补偿电流获得一个最小的相角。

## Exercises

### 1.Answer the following questions according to the text

（1）What are power frequency high voltages measurement techniques？

（2）Why is a resistance potential divider with an electrostatic voltmeter sometimes better when high precision is needed？

（3）Why are generating voltmeters safer in high DC voltage measurements？

（4）What conventional methods does measurement of high AC voltages employ？

（5）What is the impedance of the resistance $R$，at any frequency$\omega$of the AC voltage？

### 2.Translate the following sentences into Chinese according to the text

（1）In industrial testing and research laboratories，it is essential to measure the voltages accurately，ensuring perfect safety to the personnel and equipment.

（2）Therefore，even though the principles of measurements may be same，the devices and instruments for measurement of high voltages differ vastly from the low voltage devices.

（3）The net current in the meter is usually limited to one to ten microamperes for fullscale deflection.

（4）But their designs are different from those of low voltage meters， as the insulation design and source loading are the important criteria.

（5）The entire resistor unit then has to be taken as a transmission line equivalent，for calculating the effective resistance.

## 3.Translate the following paragraph into Chinese

A fault in an electrical power system is the unintentional and undesirable creation of a conducting path（a short circuit）or a blockage of current（an open circuit）.The short circuit fault is typically the most common and is usually implied when most people use the term fault.We restrict our comments to the short circuit fault.

The causes of faults include lightning， wind damage， trees falling across lines， vehicles colliding with towers or poles， birds shorting out lines， aircraft colliding with lines， vandalism， small animals entering switchgear， and line breaks due to excessive ice loading.Power system faults may be categorized as one of four types:single line-to-ground，line-toline，double line-to-ground and balanced three-phase.The first three types constitute severe unbalanced operating conditions.